



**Baker
College**

GREATNESS
IS EARNED HERE.

Electricity and Electrical Safety

Programmable Logic Controllers

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College of Information Technology and Engineering

Agenda

Electricity Review
Circuit Review

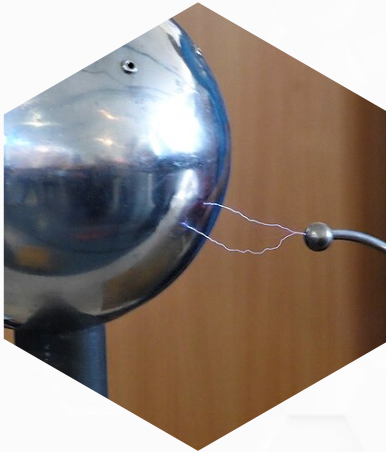
Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

Electricity Review



Electricity

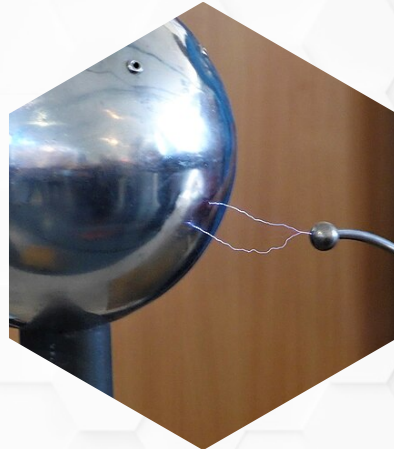


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Electricity is a form of energy resulting from the existence of charged particles (protons or electrons)

Primary Properties of Electricity

1. Voltage
2. Current
3. Resistance



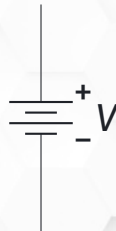
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Electricity Review

Voltage

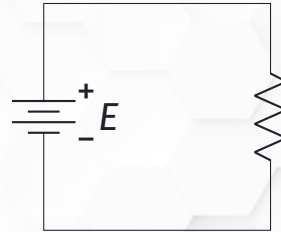
Voltage as Potential Energy

- Voltage is potential energy
- The difference between the number of electrons in the positive and negative terminals
- Potential energy + difference = *potential difference*



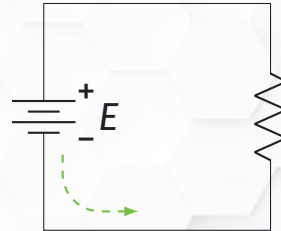
Voltage as Electromotive Force

- Voltage acts like a pressure—on its own it's a pressurized tank
- When connected to a circuit it propels electrons through the circuit
- *Electromotive force (EMF)*



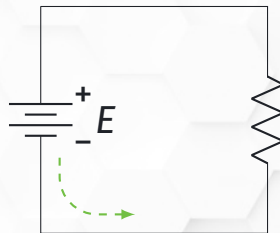
Voltage Generates Current

- Voltage pushed electrons around the circuit, creating *current*



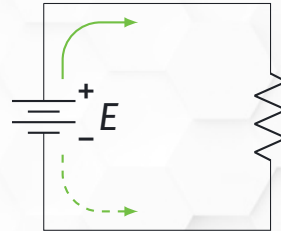
Electron Flow

- Law of physics: things move from high concentration to low concentration
- Current flows from high concentration of electrons (negative) to low concentration (positive)



Conventional Flow

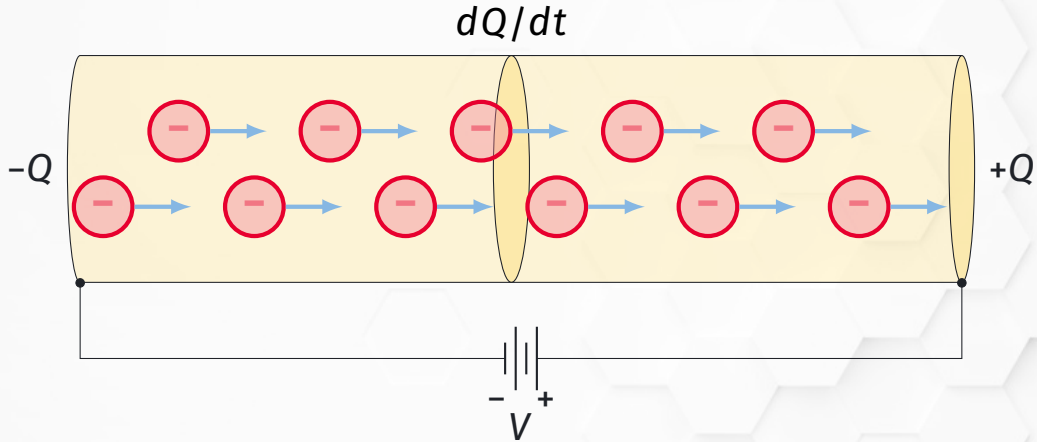
- Current is traditionally drawn *flowing from positive to negative*
- *Passive sign convention*
- Mathematically fine as long as directions and polarities are always maintained



Electricity Review

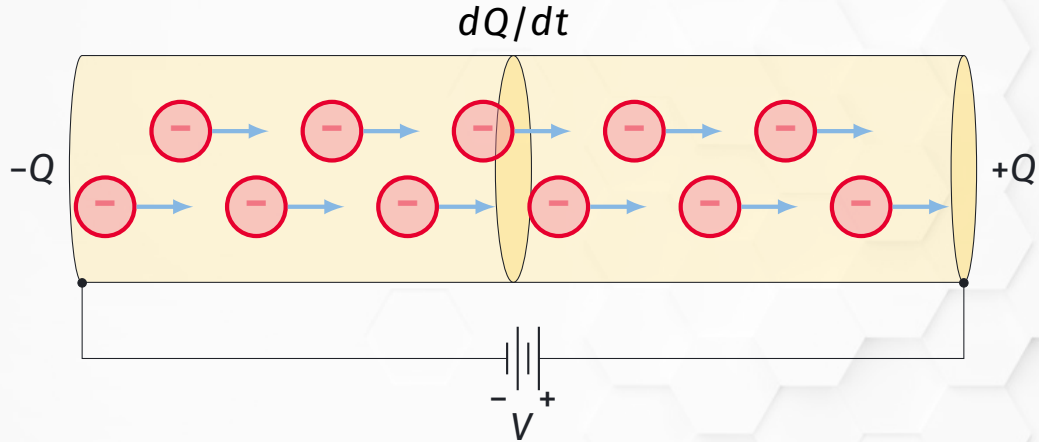
Current

Current Defined



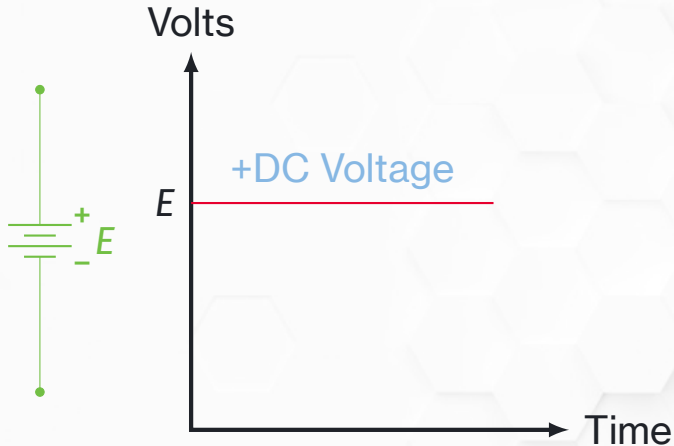
Current is the *flow rate* (speed) of electric charge through a conductor

Ampere Defined



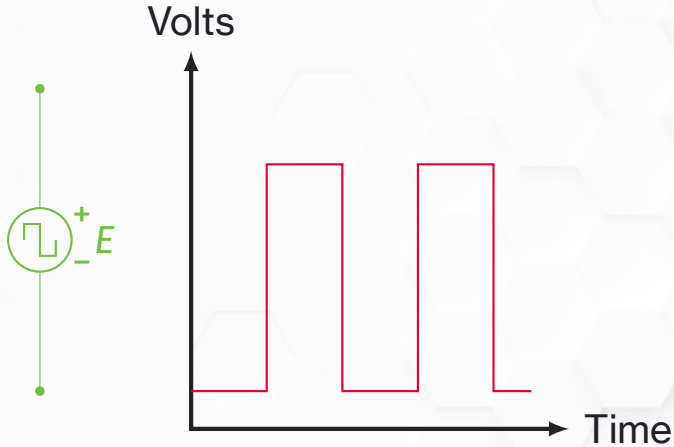
One *ampere* is 1 coulomb of charge (6.24×10^{18} electrons) flowing past a single point in 1 second

Direct Current Defined



Direct current is an electric current that maintains its polarity (flows in only one direction)

Pulsed DC



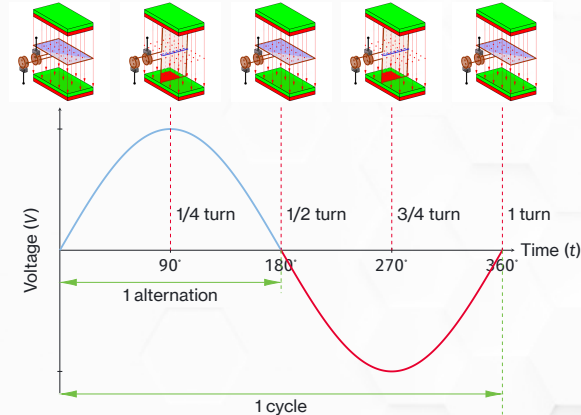
Direct current **does not** need to be a steady voltage – it can be *pulsed*. As long as it maintains polarity it is still DC.

Alternating Current Defined



Alternating current is an electric current that reverses polarity. It **does not** need to be a sine wave, but the sine wave is the most common.

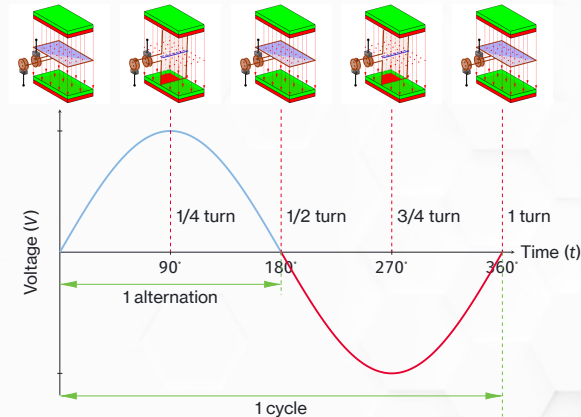
AC Voltage - Sine Wave



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The sine wave is caused by the physical layout of an AC generator – as the coil spins inside the magnetic field it cuts the field in different directions, leading to a reverse in polarity

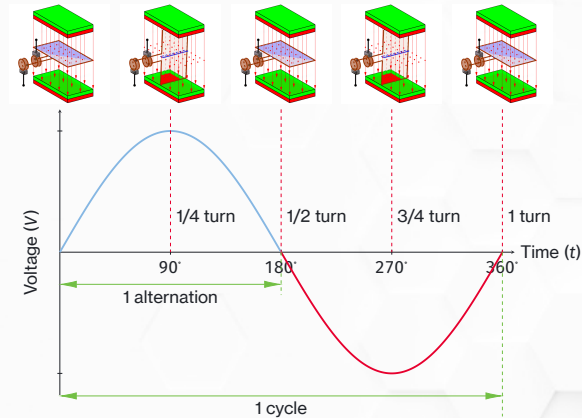
AC Voltage - Alternation



Attribution: Adapted from "Electric Generator Animation" (frames 5 and 17 extracted) by MikeRun, CC BY-SA 4.0, via Wikimedia Commons.

An *alternation* is 1/2 of a rotation of the coil—the point at which the polarity changes

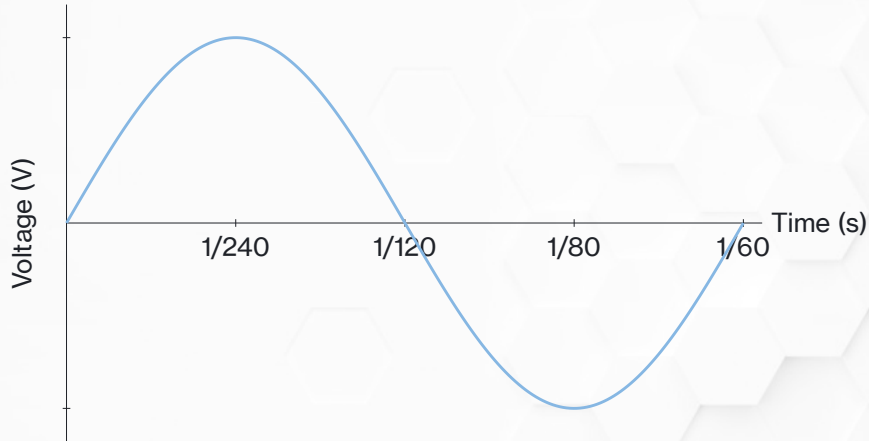
AC Voltage - Cycle



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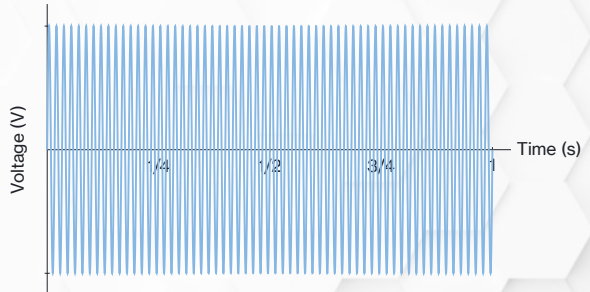
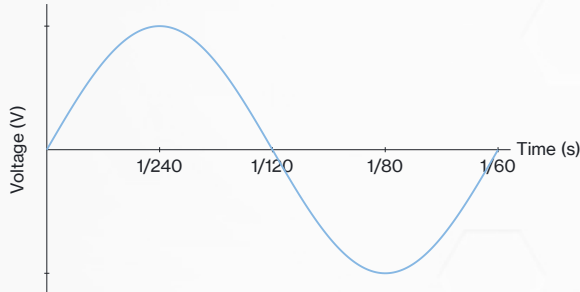
A cycle is a full rotation of the coil

AC Voltage - Frequency



The frequency is the number of cycles per second (Hz)–the U.S. standard is 60 Hz (60 cycles per second)

Visualizing Frequency

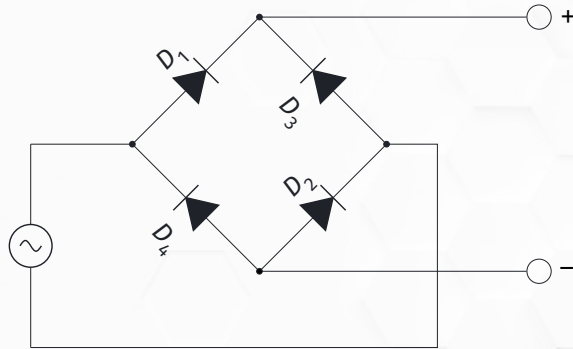


We often use one cycle (left) to visualize the curve, but remember that that one cycle is happening 60 times every second (right)

Electricity Review

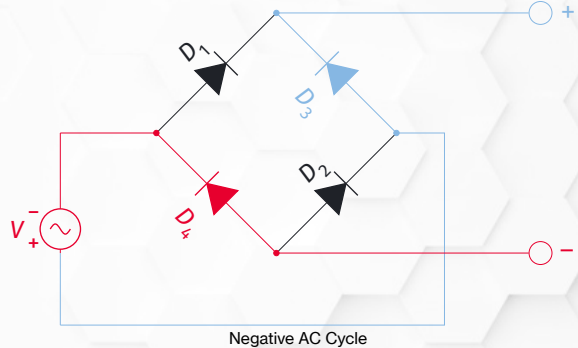
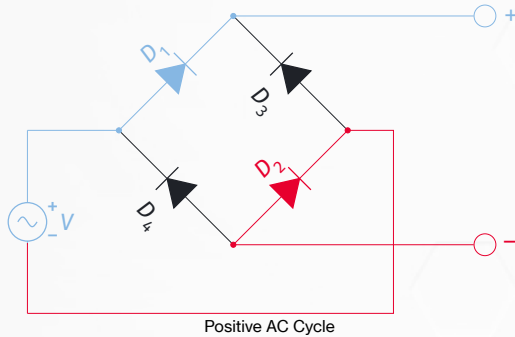
Converting AC to DC

Bridge Rectifier



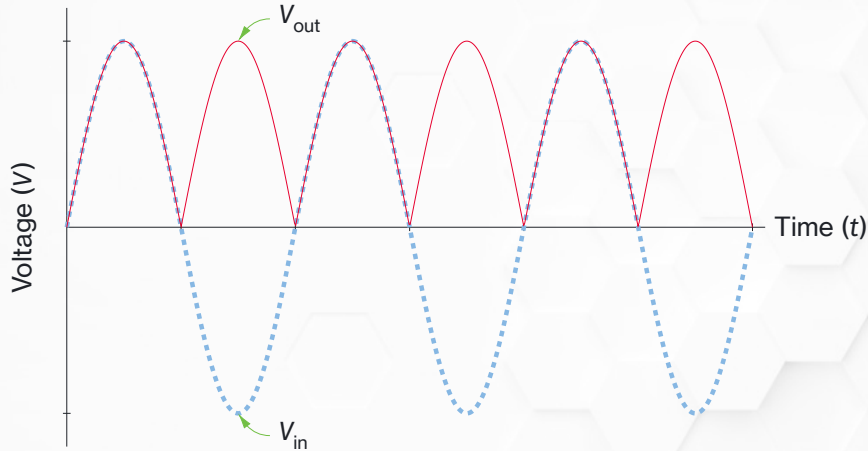
A *bridge rectifier* is commonly used in controls applications to convert AC to DC

Bridge Rectifier - Active



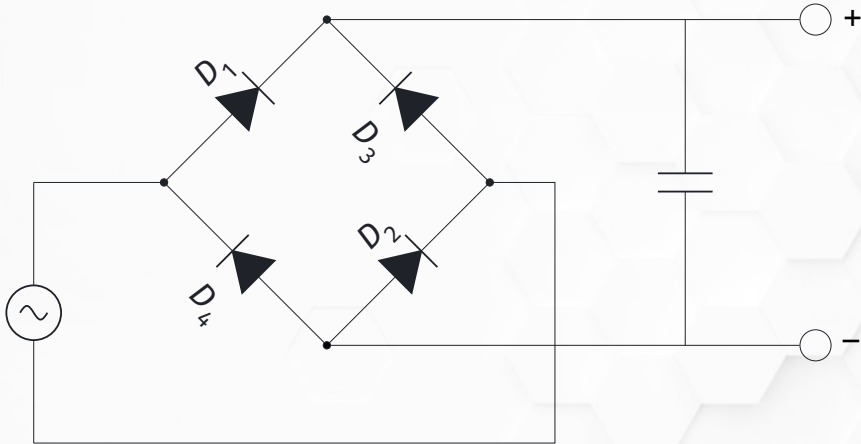
As the polarity of the AC voltages changes the the rectifier forces the output to the same polarity

Rectified Alternating Current



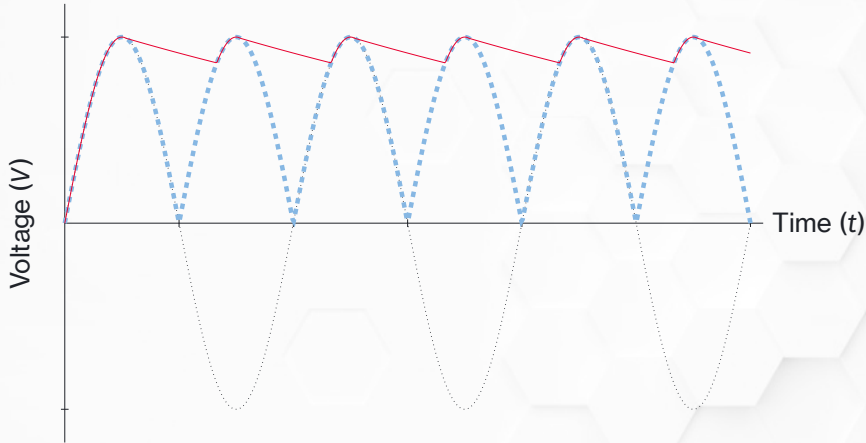
The rectifier redirects current so the output is always the same polarity, but it *does not* affect the voltage value—the resulting wave is DC, but not very useful DC

Bridge Rectifier - Smoothed



Smoothing capacitors are added to help maintain the voltage level between cycles, which helps keep the DC at a more consistent value

Rectified and Smoothed Alternating Current

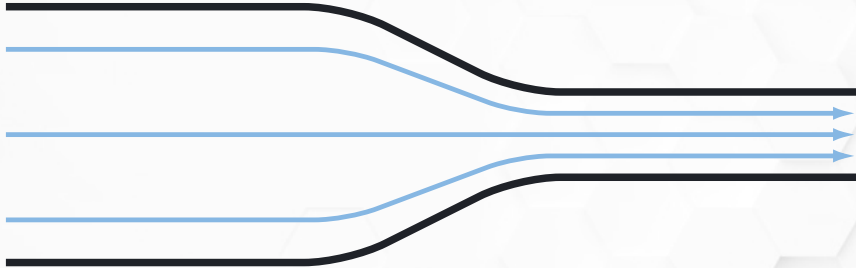


The result is a fairly smooth DC wave—the *ripple* will always be there, but can be tweaked by changing the capacitor value

Electricity Review

Resistance

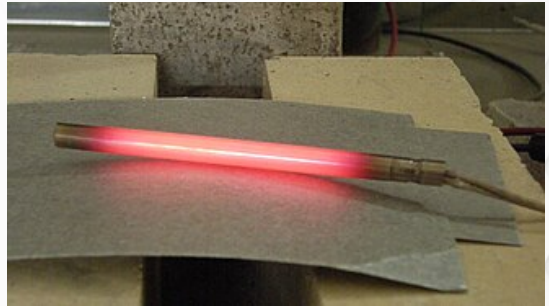
Resistance Defined



Resistance is the opposition to electron flow—like a restriction in a water pipe

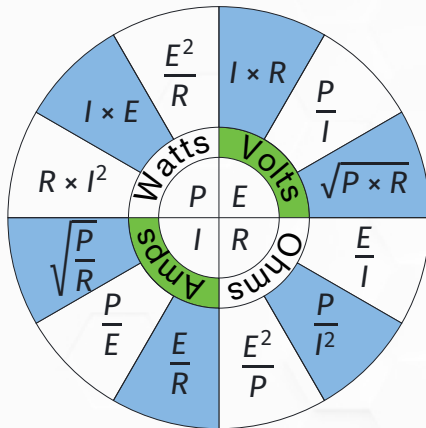
Resistance Causes Heat

- *ALL* components in a circuit have resistance—including wires
- Current through a resistance *always* comes with heat
- Unless we're actually building a heater, this is *dissipated* (lost) energy



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Ohm's Law

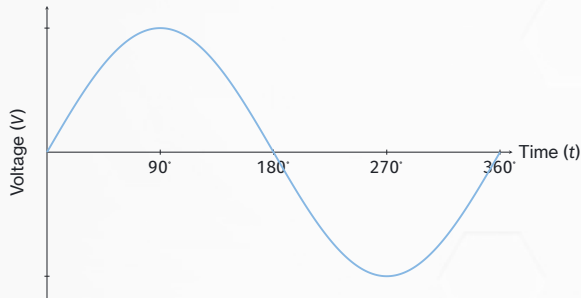


Ohm's Law relates current, voltage, and resistance. Combined with Watt's Law to relate power (heat) we can solve all aspects of a circuit.

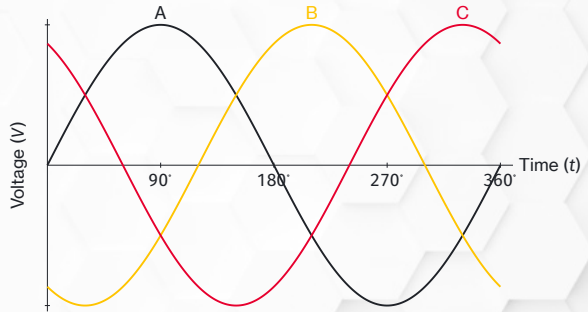
Electricity Review

Voltage Phases

AC Voltage - Phases

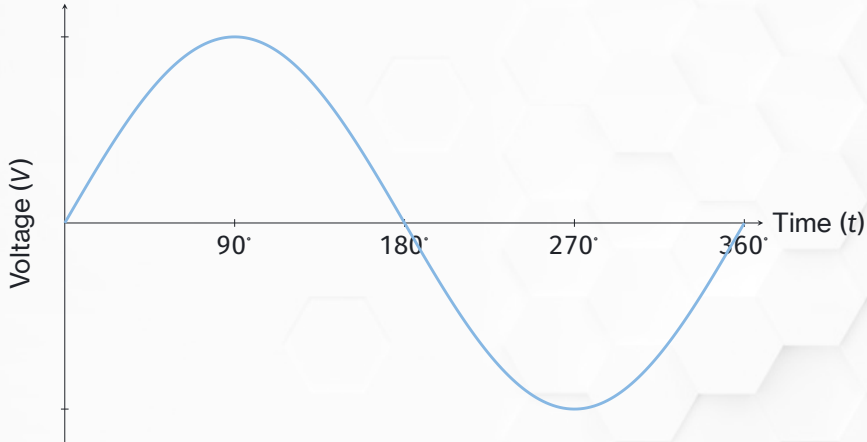


1 ϕ is created with a single alternator coil



3 ϕ is created with 3 coils physically placed 120° apart

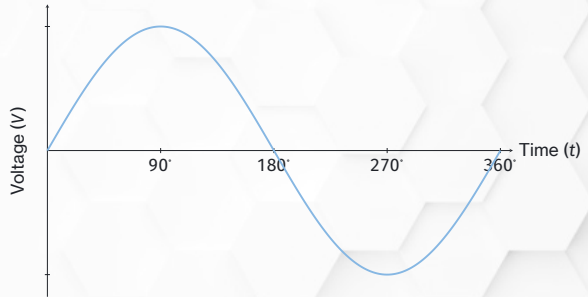
Single Phase - Zero Voltage (1 of 5)



The instantaneous voltage value of 1 ϕ crosses 0 twice every cycle

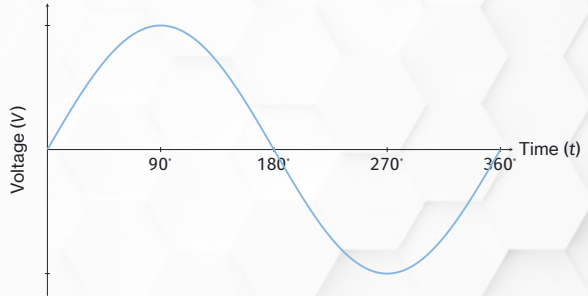
Single Phase - Zero Voltage (2 of 5)

- What happens when voltage drops to 0?



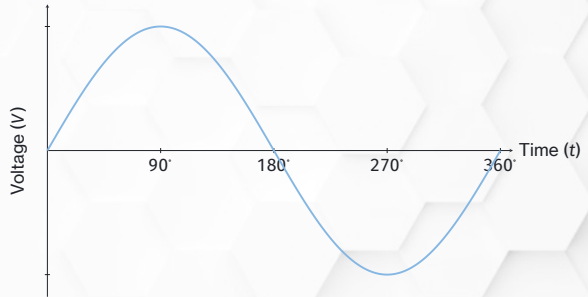
Single Phase - Zero Voltage (3 of 5)

$$p(t) = v(t) \cdot i(t)$$
$$v(t) = 0 \rightarrow p(t) = 0$$

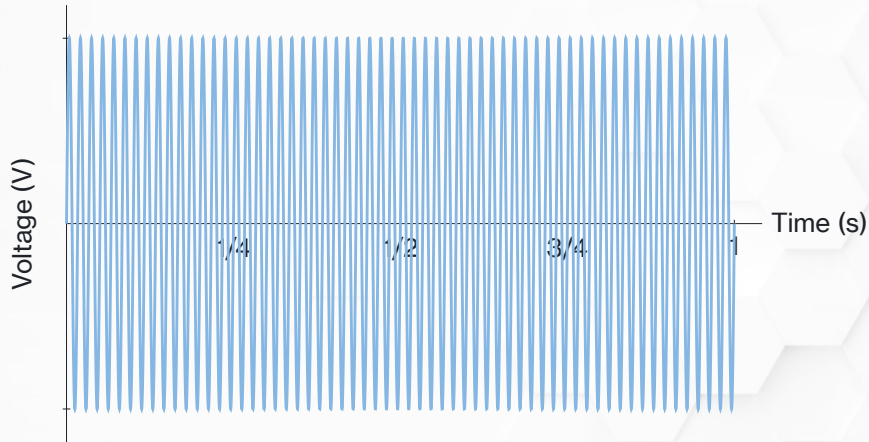


Single Phase - Zero Voltage (4 of 5)

- Instantaneous power drops to zero twice every cycle
- Requires higher peak current I_{peak} to deliver equal average power P_{avg}
- Causes mechanical vibration and pulsating torque in heavy equipment

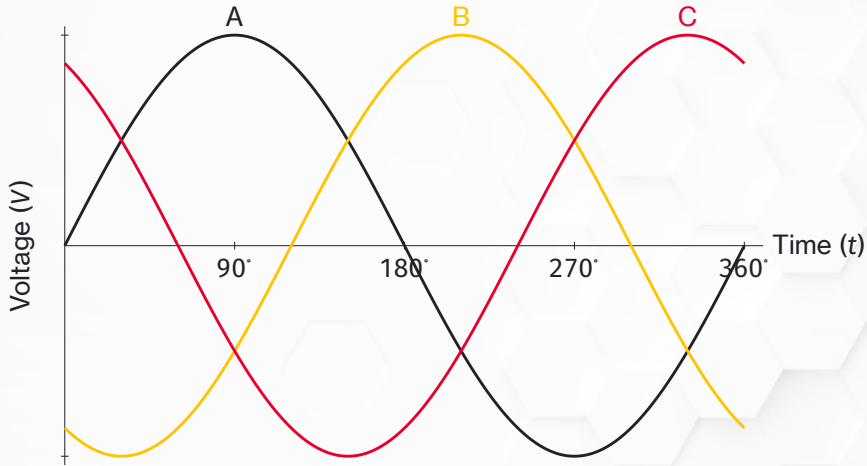


Single Phase - Zero Voltage (5 of 5)



This isn't a rare occurrence—at 60 Hz the power is dropping to zero 120 times every second

Three Phase Power



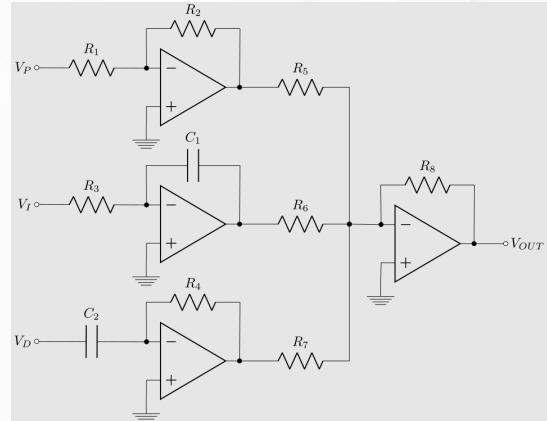
With a 3 ϕ power supply the effective voltage is *never* zero—less average current required to maintain power and no pulsing

Circuit Review



Electric Circuit Defined

- A closed, complete loop of conductive components that allow current to flow from a source, through a device, and back to the source

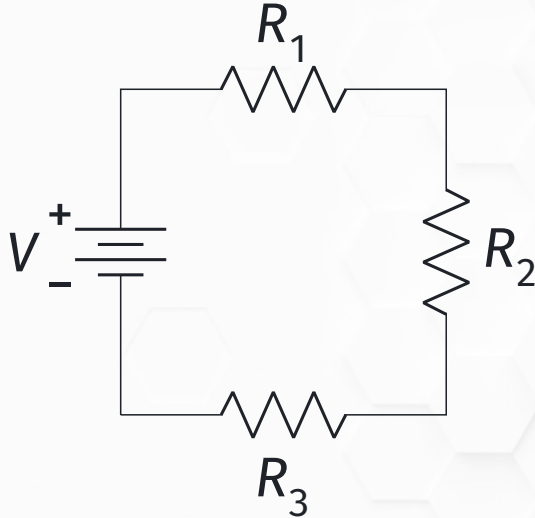


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Circuit Review

Series Circuits

Series Circuit Defined



Ladder Logic Test

